

Chemistry

Time Allowed : 1 hour _____ **Maximum Marks : 100**

Please read the instructions carefully. You will be allotted 5 minutes specifically for this purpose.

Instructions

A. General

1. Blank papers, clipboards, log tables, slide rules, calculators, cellular phones, pagers, and electronic gadgets in any form are not allowed.
2. Do not break the seals of the question-paper booklet before instructed to do so by the invigilators.

B. Question paper format and Marking Scheme :

1. Paper contains 10 questions. Each question in this section carries 10 marks.

- Q1:** 75.2 g of (phenol) is dissolved in 1 kg of solvent of = 14 K molality . If depression in freezing point is 7K, Calculate % of phenol that dimerises.
- Q2:** A solution of a non-volatile solute in water freezes at -0.30°C . The vapour pressure of pure water at 298 K is 23.51 mm Hg and K_f for water is 1.86 degree/molal. Calculate the vapour pressure of this solution at 298 K.
- Q3:** 2 g of benzoic acid dissolved in 25 g of benzene shows a depression in freezing point equal to 1.62 K. Molal depression constant of benzene is 4.9 K kg. What is the percentage association of acid if it forms double molecule in solution?
- Q4:** 1.4 g of acetone dissolved in 100 g of benzene gave a solution which freezes at 277.12 K. Pure benzene freezed at 278.4 K. 2.8 g of a solid (A) dissolved in 100 g of benzene gave a solution which froze at 277.76 K. Calculate the molecular weight of (A).
- Q5:** The vapour pressure of pure benzene at a certain temperature is 640 mm of Hg. A non-volatile non-electrolyte solid weighing 2.175 g is added 39.0 g of benzene. The vapour pressure of the solution is 600 mm of Hg. What is the molecular weight of solid substance?
- Q6:** The formula weight of an acid is 82.0. 100ml of a solution of this acid containing 39.0 g of the acid per litre were completely neutralized by 95.0ml of aqueous NaOH containing 40.0 g of NaOH per litre. What is the basicity of the acid?
- Q7:** A sample of hard water contains 96 ppm. of SO_4^{2-} and 183 ppm of HCO_3^- . With Ca^{2+} as the only cation. How many moles of CaO will be required to remove HCO_3^- from 1000 kg of this water? If 1000 kg of this water is treated with the amount of CaO calculated above, what will be the concentration (in ppm) of residual Ca^{2+} ions in one litre of the treated water are completely exchanged with hydrogen ions, what will be its pH (One ppm means one part of the substance in one million part of water, weight/weights)?
- Q8:** 1 g charcoal is placed in 100 mL of 0.5 M CH_3COOH to form an adsorbed mono-layer of acetic acid molecule and thereby the molarity of CH_3COOH reduces to 0.49. Calculate the surface area of charcoal adsorbed by each molecule of acetic acid. Surface area of charcoal = 3.01 .
- Q9:** n- butane is produced by the monobromination of ethane following by Wurtz reaction. Calculate the volume of ethane at NTP to produce 55 g n-butane if the bromination takes place with 90% yield and the Wurtz reaction with 85% yield.
- Q10:** A mixture of HCOOH and $\text{H}_2\text{C}_2\text{O}_4$ is heated with conc. H_2SO_4 . The gas produced is collected and on treating with KOH solution the volume of the gas decreases by $\frac{1}{6}$ th. Calculate molar ratio of two acids in original mixture.